Kai Gu

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Education

Bocconi University — **Milan, Italy**
PhD in Marketing — 2026 – Present

Bocconi University — **Milan, Italy**
MSc in Data Science — 2024 – 2026

Zhejiang University — **Hangzhou, China**
BEcon in Economics — 2020 – 2024

Research Interests

Substantive: Economics of AI, Economics of Digitization & Platforms, Media Economics
Methodological: Causal Inference, Computational Social Sciences, Multimodal Data

Working Papers

From Directory to Oracle: How AI Search Reshapes the Web's Attention Economy
Qiaoni Shi, Kai Zhu, and **Kai Gu**

Presented at: Bocconi Marketing Research Camp, May 2026; ISMS Marketing Science Conference, Lisbon, Portugal, June 2026

For two decades, the web's attention economy has rested on an implicit bargain: search engines help users find information and route traffic to the websites that produce it. AI search threatens to break this bargain by serving as an oracle that delivers synthesised answers rather than a directory that distributes attention. Using URL-level clickstream data on 45,386 US desktop panelists over October 2024–July 2025, we provide the first within-user, multi-shock empirical characterisation of how AI search reshapes web traffic. Four facts: (i) ChatGPT Search adoption was rapid and shock-driven, with three access expansions lifting search-enabled users ∼1.7× to ∼3.3×; (ii) AI search is absorptive—only 5.4% of ChatGPT conversations produce any outbound click vs. 33.1% of Google queries, and 67% of ChatGPT users never click out at all (vs. 11% for Google), with a within-user referral-rate gap of ∼30 percentage points; (iii) AI search referrals are systematically tilted toward reference, developer, and academic destinations and away from ad-supported and community content, and within a user-week ChatGPT reaches only ∼2 unique domains vs. Google's ∼14; (iv) exploiting two ChatGPT Search rollouts in a stacked event-study design, AI-search adoption causes a ∼9–11% reduction in Google queries, concentrating in informational destinations (Wikipedia, GitHub, NYT) while sparing transactional ones (Amazon).

Governing AI Content Tools: The Efficiency–Diversity Trade-off in Platform Knowledge Ecosystems
Kai Zhu, **Kai Gu**, and Lorenzo Lupo

Presented at: 13th Wiki Workshop (Online), March 2026

Knowledge platforms increasingly deploy AI tools to accelerate content production, yet the ecosystem-level consequences of these decisions remain poorly understood. We audit Wikipedia's machine translation (MT) deployment—one of the earliest large-scale platform adoptions of AI content tools—using over half a million matched article pairs across ten language editions. We document an efficiency–diversity trade-off: MT-assisted articles provide 50% greater information volume to target-language readers, yet total non-redundant information in the ecosystem is virtually unchanged (+2.6% moving from 0% to 100% MT adoption), because MT substitutes translated source material for locally originated content rather than expanding the information frontier. MT-assisted articles exhibit substantially lower semantic divergence from English sources (Semantic Divergence Index: 0.16 vs. 0.43), target-unique content falls by 48%, and the absent material is disproportionately culturally anchored—local context, regional perspectives, and cultural details that human editors naturally add. A difference-in-differences design exploiting a 2019 NMT quality shock shows the gap is causal: improved MT technology widens the diversity deficit rather than closing it. A bootstrap simulation maps the efficiency–diversity frontier across adoption rates, yielding trade-off ratios that serve as actionable governance instruments for platform managers.

Conference Presentations

ISMS Marketing Science Conference, Lisbon, Portugal	June 2026
Bocconi Marketing Research Camp	May 2026
13th Wiki Workshop (Online)	March 2026

Skills

Languages: English, Mandarin
Programming: R, Python, LaTeX, Markdown